

Chapter 2 (part I)

Bivariate Relationship and Two-way Table

2.1 Relationships

2.2 Scatterplots

2.3 Correlation

2.6 Data Analysis for Two-Way Tables

Associations Between Variables

2

Two variables measured on the same cases are **associated** if knowing the value of one of the variables tells you something that you would not otherwise know about the value of the other variable.

When you examine the relationship between two variables, a new question becomes important: *Is your purpose simply to explore the nature of the relationship, or do you wish to show that one of the variables can explain variation in the other?* E.g. physical activity level and weight.

A **response variable** measures an outcome of a study. An **explanatory variable** may help explain or predict variation in the response variable.

Scatterplot 1

The most useful graph for displaying the relationship between two quantitative variables is a **scatterplot**.

A **scatterplot** shows the relationship between two quantitative variables measured on the same cases. The values of one variable appear on the horizontal axis, and the values of the other variable appear on the vertical axis. *Each case corresponds to one point on the graph.*

How to Make a Scatterplot

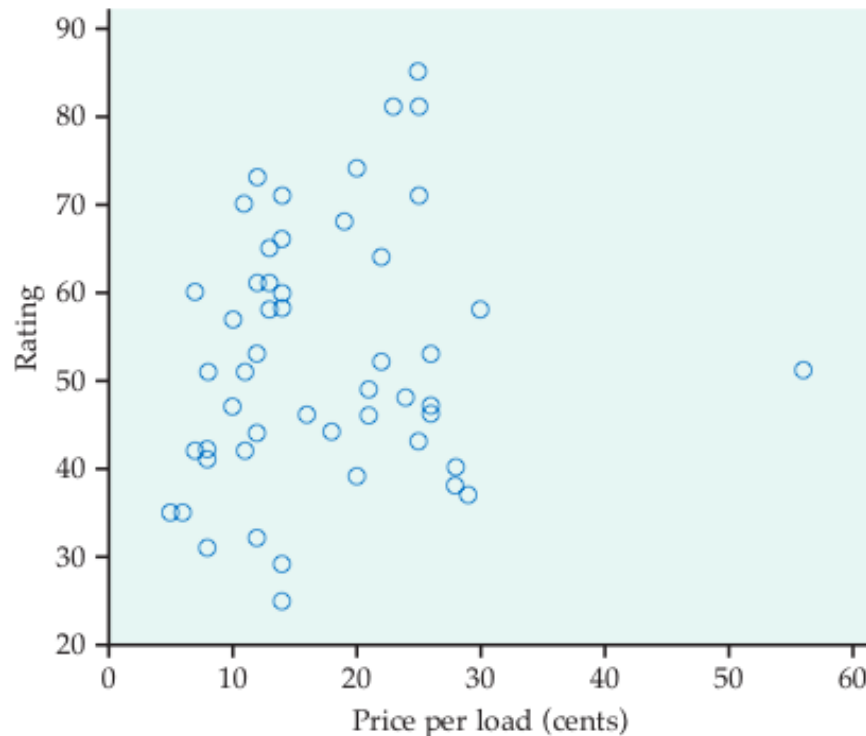
1. Decide which variable should go on each axis. If a distinction exists, *plot the explanatory variable on the x axis and the response variable on the y axis.*
2. Label and scale your axes.
3. Plot individual data values.

Scatterplot 2

4

Example: Make a scatterplot of the relationship between price and rating for laundry detergents.

Price per load (cents)	14	14	14	12	10	8	...	12
Rating	60	58	60	53	57	41	...	32



Interpreting Scatterplots 1

How to Examine a Scatterplot

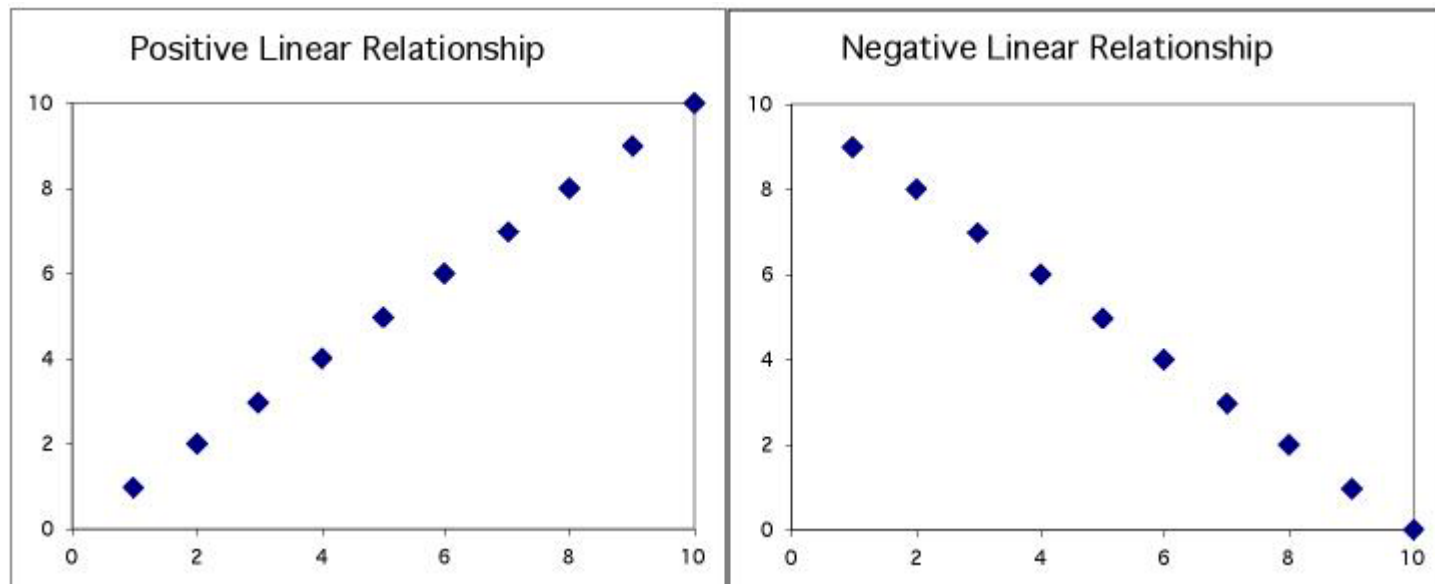
As in any graph of data, look for the *overall pattern* and for striking *deviations* from that pattern.

- You can describe the overall pattern of a scatterplot by the **form**, **direction**, and **strength** of the relationship.
- An important kind of departure is an **outlier**, an observation that falls outside the overall pattern of the relationship.

Interpreting Scatterplots 2

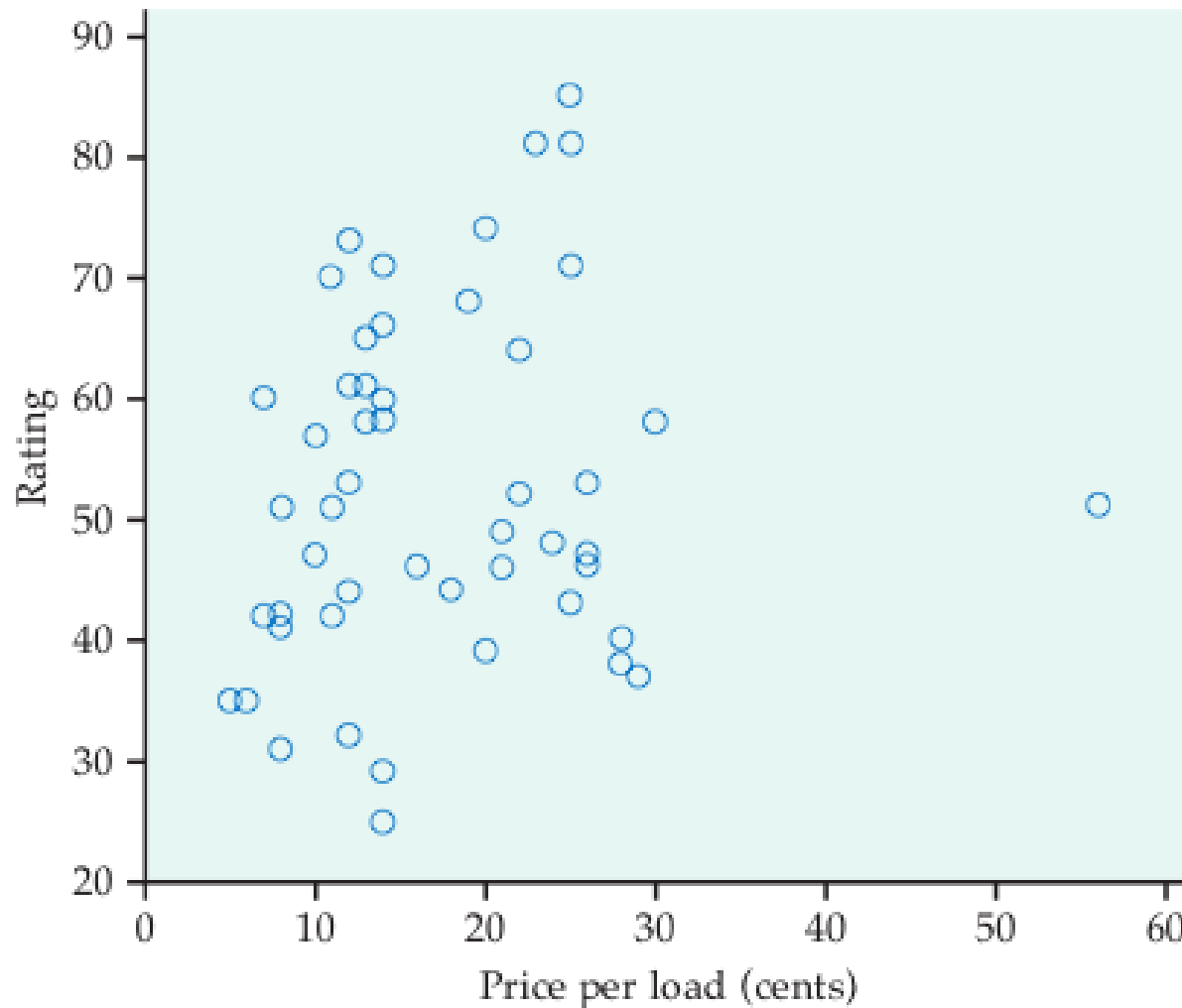
Two variables are **positively associated** when above-average values of one tend to accompany above-average values of the other and when below-average values also tend to occur together.

Two variables are **negatively associated** when above-average values of one tend to accompany below-average values of the other and vice versa.



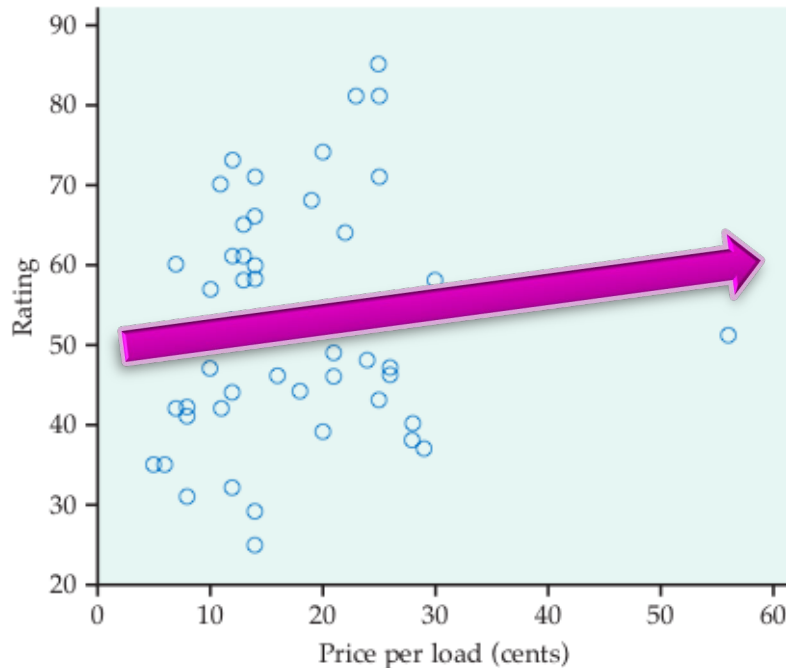
Interpreting Scatterplots 3

7



Interpreting Scatterplots 4

8



Outlier

- ✓ There is one possible unusual observation—the detergent with the price per load of 56 cents seems to have a lower rating than one might expect **if higher price means a higher rated detergent.**

Strength

Direction

Form

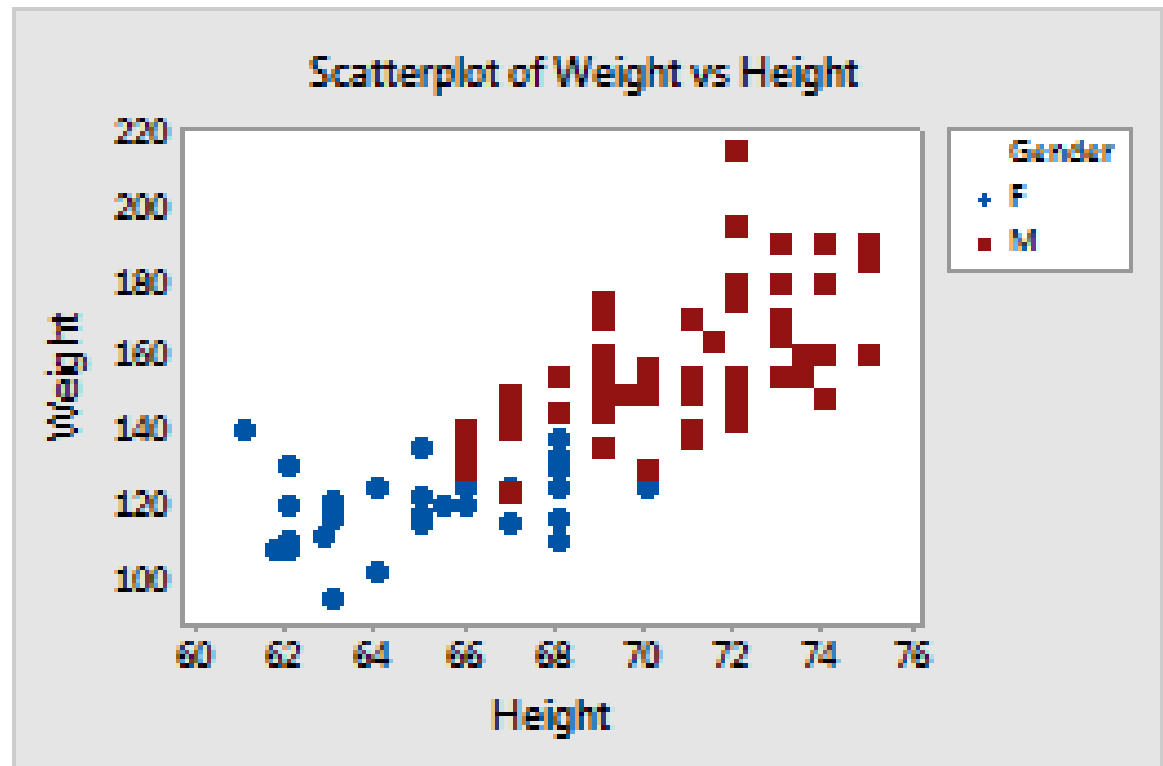
- ✓ There is a **moderately weak, positive, linear** relationship between price per load and detergent rating.
- ✓ It appears that more expensive detergents are rated slightly higher, with possibly one exception.

Adding Categorical Variables

10

- Consider the relationship between height and weight.

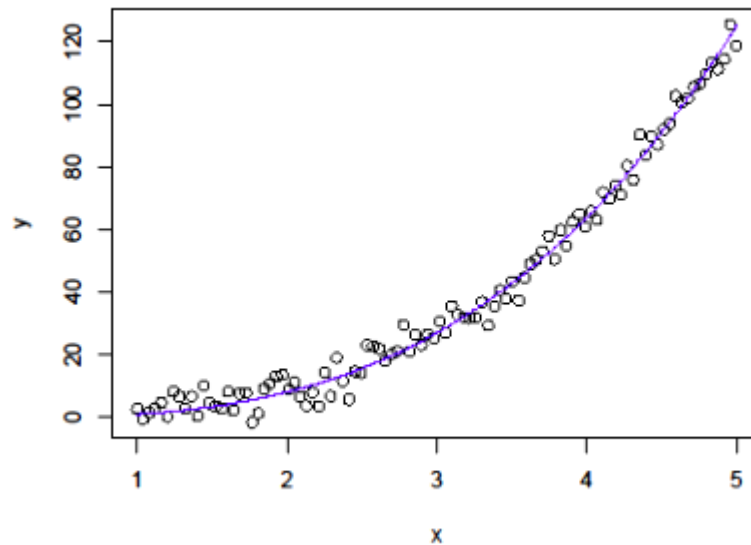
To add a *categorical variable*, use a different plot color or symbol for each category.



Nonlinear Relationships

10

- There are other **forms** of relationships besides linear. The scatterplot below is an example of a **nonlinear form**.
- Note the curvature in the relationship between x and y .



Log Transformation

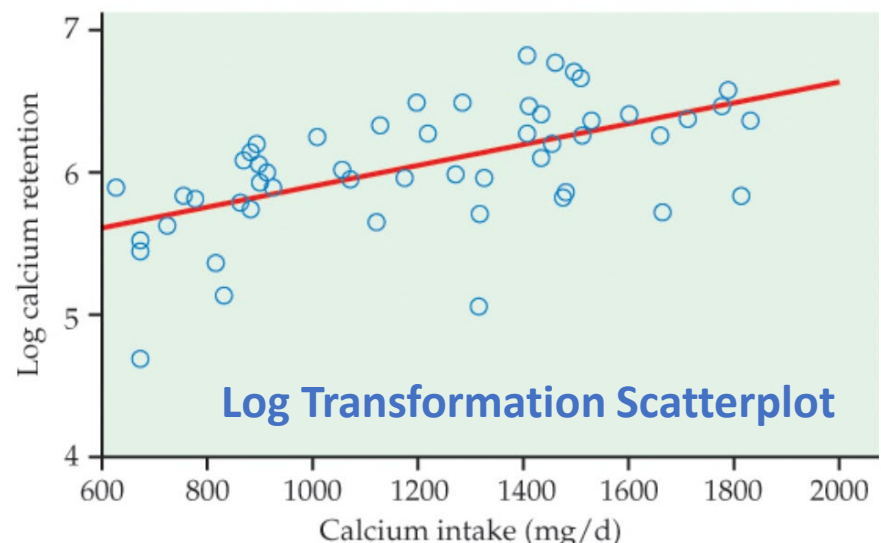
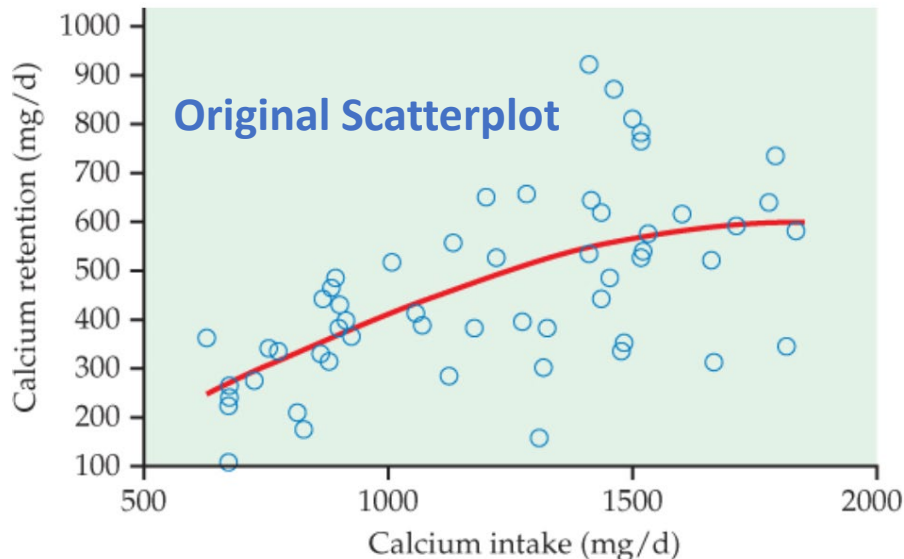
11

Sometimes, the pattern that best describes data in a scatterplot is a curve other than a line. In this case, it is often useful to apply a **transformation** to the data that renders the relationship approximately linear.

The **log transformation**:

- can be used only for variables whose values are positive.
- A common choice is the **natural logarithm**.

Example: Consider the following two scatterplots concerning calcium intake versus calcium retention.



2.3 Correlation

- Correlation coefficient r
- Properties of r

Measuring Linear Association 1

A scatterplot displays the strength, direction, and form of the relationship between **two quantitative variables**. **Linear** relationships are important because a straight line is a simple pattern that is quite common.

Our eyes are not always good judges of how strong a relationship is. Therefore, we use a numerical measure to supplement our scatterplot and help us quantify the strength of the linear relationship.

The **correlation (coefficient) r** measures the **direction** and **strength** of the **linear** relationship between two **quantitative** variables. Using the notation explained on pp. 103–104 in the text:

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Measuring Linear Association 2

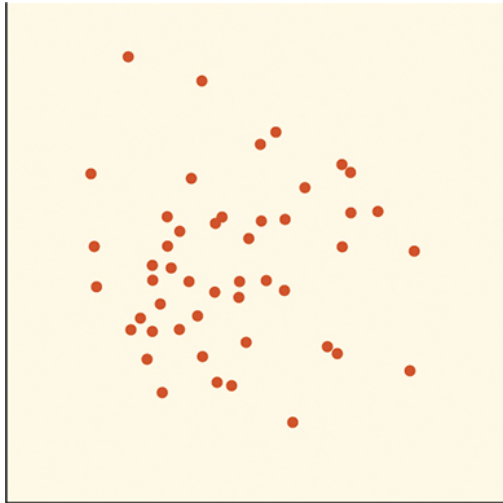
We say a **linear** relationship is **strong** if the points lie close to a straight line and **weak** if they are widely scattered about a line. The following facts about r help us further interpret the strength of the linear relationship.

Properties of Correlation

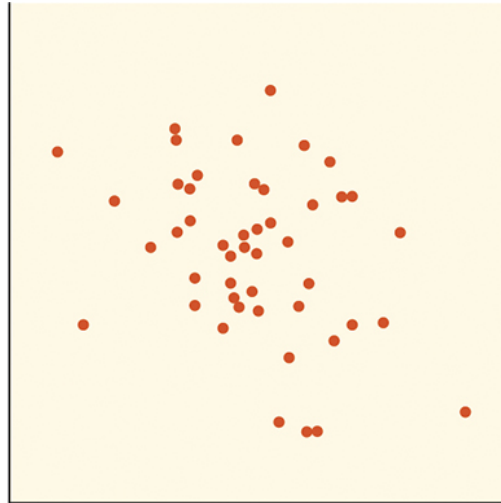
- r is always a number between -1 and 1 .
- $r > 0$ indicates a positive association.
- $r < 0$ indicates a negative association.
- Values of r near 0 indicate little or even no linear association.
- The strength of the linear relationship increases as r moves away from 0 toward -1 or 1 .
- The extreme values $r = -1$ and $r = 1$ occur only in the case of a perfect linear relationship.

Correlation

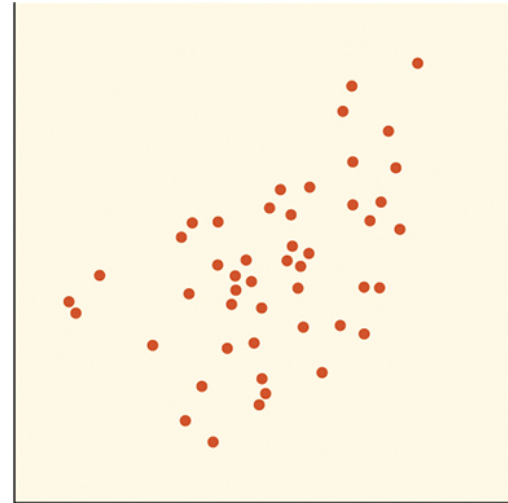
15



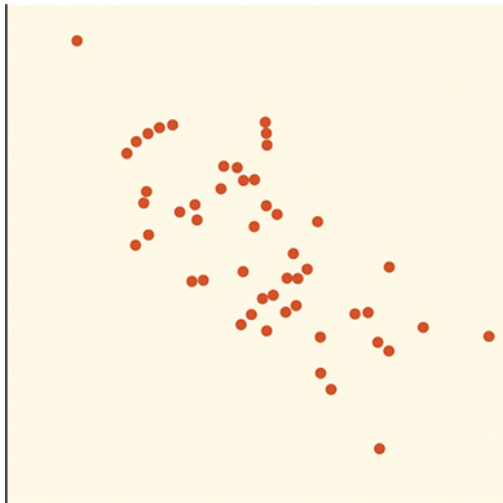
Correlation $r = 0$



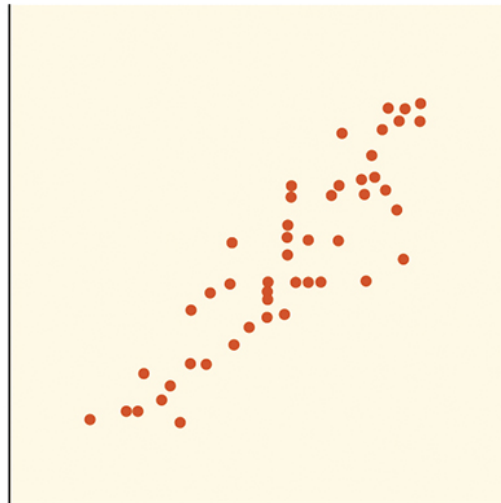
Correlation $r = -0.3$



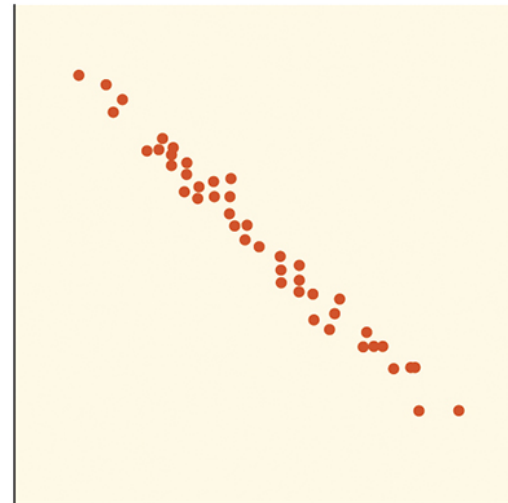
Correlation $r = 0.5$



Correlation $r = -0.7$



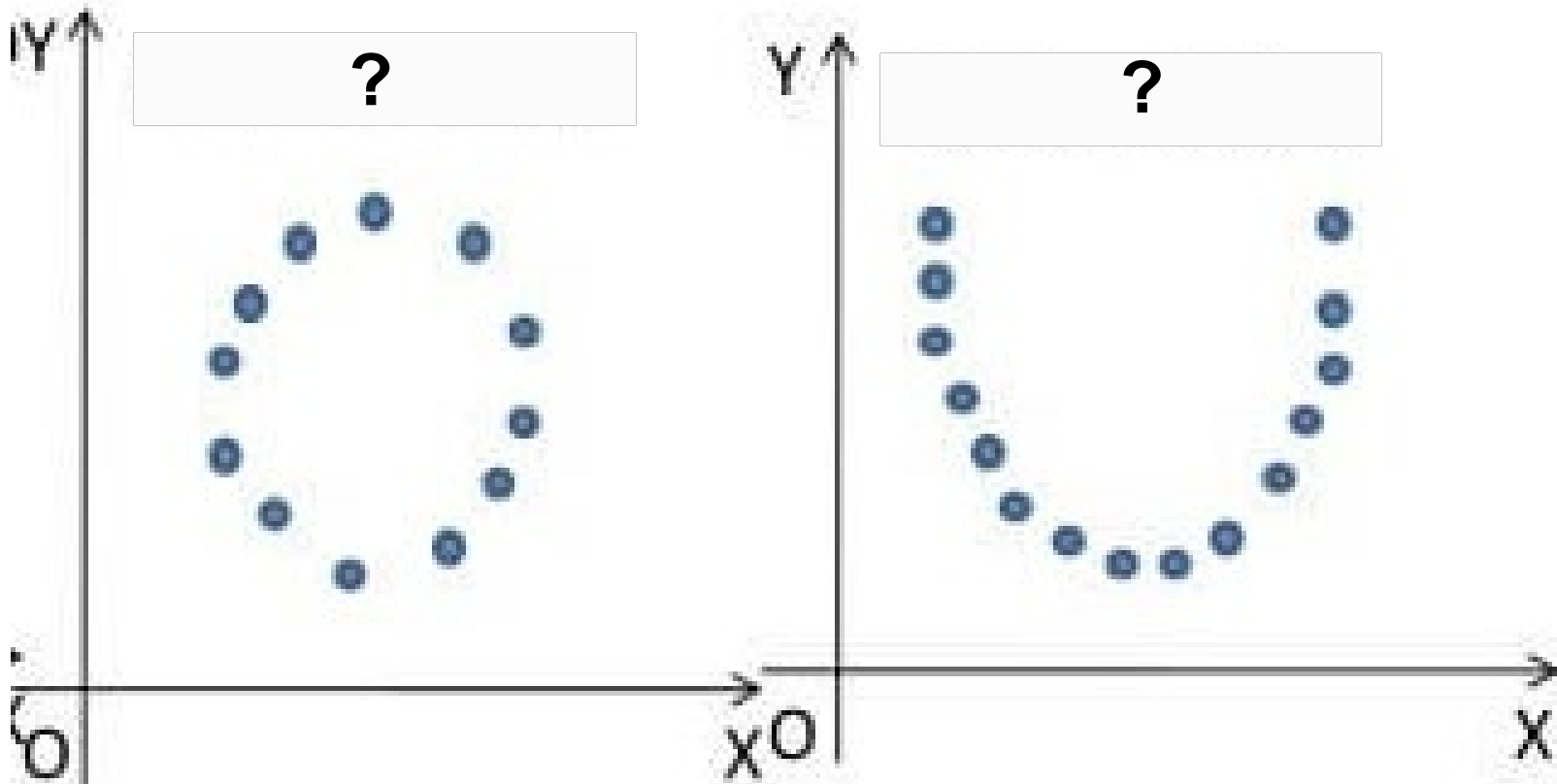
Correlation $r = 0.9$



Correlation $r = -0.99$

Correlation

16



Properties of Correlation

17

1. Correlation makes no distinction between explanatory and response variables.
2. r has *no units* and *does not change when we change the units of measurement of x , y , or both (why?)*.
3. Positive r indicates positive association between the variables, and negative r indicates negative association.
4. The correlation r is always a number between -1 and 1 .

Cautions:

- Correlation requires that both *variables be quantitative*.
- Correlation *does not describe curved relationships* between variables, no matter how strong the relationship is.
- The correlation r is not resistant; it can be strongly affected by a few outlying observations.
- *Correlation is not a complete summary of two-variable data.*

2.6 Data Analysis for Two-Way Tables

- The two-way table
- Marginal distributions
- Conditional distributions
- Simpson's paradox

Categorical Variables

Recall that categorical variables place individuals into one of several groups or categories.

- The values of a categorical variable are labels for the different categories (i.e., male and female for “sex” variable).
- The distribution of a categorical variable lists the **count** or **percent** of individuals who fall into each category.

A **two-way table** describes two categorical variables, organizing counts according to a **row variable** and a **column variable**. Each combination of values for these two variables is called a **cell**.

The Two-Way Table

20

Young adults by gender and chance of getting rich by age 30

	Female	Male	Total
Almost no chance	96	98	194
Some chance, but probably not	426	286	712
A 50–50 chance	696	720	1416
A good chance	663	758	1421
Almost certain	486	597	1083
Total	2367	2459	4826

What are the variables described by this two-way table?

How many young adults were surveyed?

Marginal Distribution 1

The **marginal distribution** of one variable in a two-way is the distribution of that variable ignoring the other variable.

Note: Percentages are often more informative than counts, especially when comparing groups of different sizes.

To examine a marginal distribution:

1. Divide each row (or column) total by the grand total to obtain the marginal distribution in percentages.
2. Make a graph to display the marginal distribution.

Marginal Distribution 2

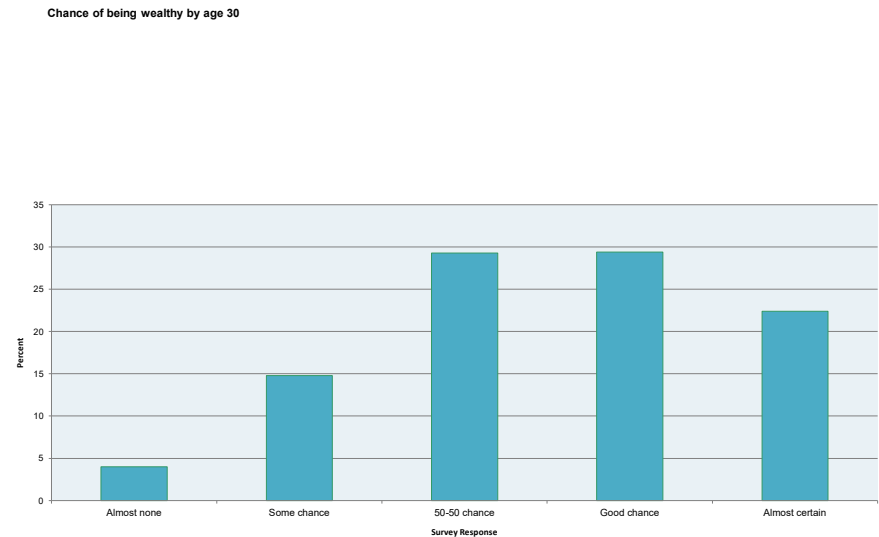
22

Young adults by gender and chance of getting rich

	Female	Male	Total
Almost no chance	96	98	194
Some chance, but probably not	426	286	712
A 50–50 chance	696	720	1416
A good chance	663	758	1421
Almost certain	486	597	1083
Total	2367	2459	4826

Examine the **marginal distribution** of **chance of getting rich**.

Response	Percent
Almost no chance	$194/4826 = 4.0\%$
Some chance	$712/4826 = 14.8\%$
A 50–50 chance	$1416/4826 = 29.3\%$
A good chance	$1421/4826 = 29.4\%$
Almost certain	$1083/4826 = 22.4\%$



Conditional Distribution 1

23

Marginal distributions tell us nothing about the relationship between two variables. For that, we need to explore the conditional distributions of the variables.

A **conditional distribution** of a variable describes the distribution of that variable among individuals who **have a specific value of another variable**.

To examine or compare conditional distributions:

1. Select the row(s) or column(s) of interest.
2. Use the data in the table to calculate the conditional distribution (**in percent**) of the row(s) or column(s). Note that the denominator is the total for the group being conditioned on, not the grand total.
3. Make a graph to display the conditional distribution.
 - Use a **side-by-side bar graph** or **segmented bar graph** to compare distributions.

Conditional Distribution 2

24

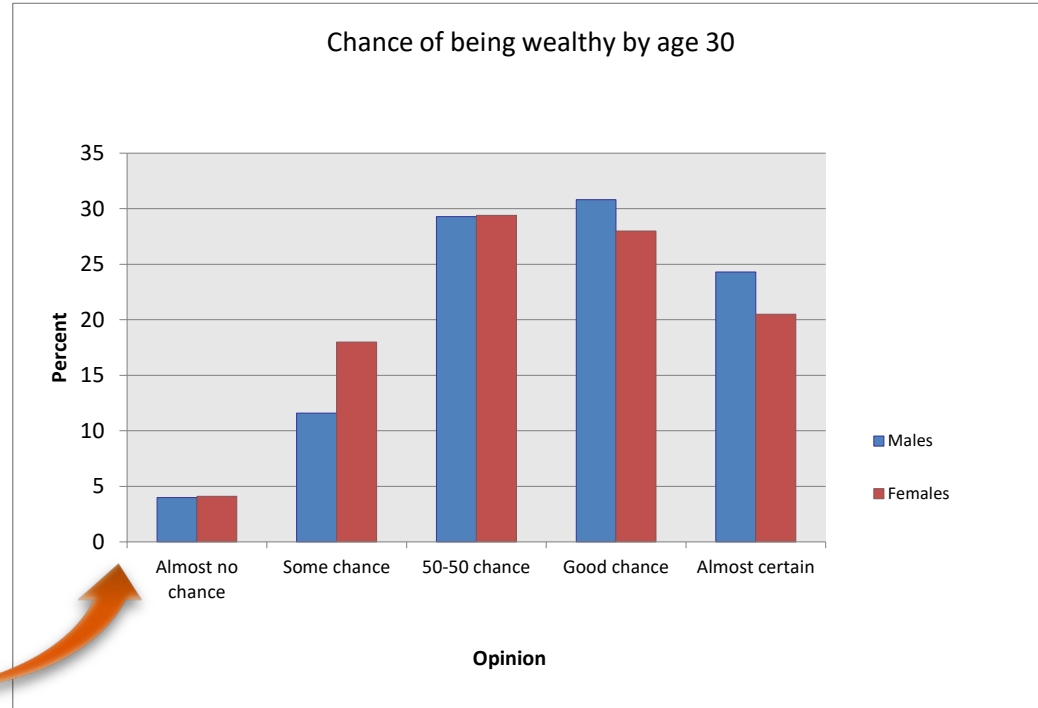
Young adults by gender and opinion of getting rich

	Female	Male	Total
Almost no chance	96	98	194
Some chance, but probably not	426	286	712
A 50–50 chance	696	720	1416
A good chance	663	758	1421
Almost certain	486	597	1083
Total	2367	2459	4826

Calculate the conditional distribution of opinion among males.

Examine the relationship between gender and opinion.

Response	Male	Female
Almost no chance	$98/2459 = 4.0\%$	$96/2367 = 4.1\%$
Some chance	$286/2459 = 11.6\%$	$426/2367 = 18.0\%$
A 50–50 chance	$720/2459 = 29.3\%$	$696/2367 = 29.4\%$
A good chance	$758/2459 = 30.8\%$	$663/2367 = 28.0\%$
Almost certain	$597/2459 = 24.3\%$	$486/2367 = 20.5\%$



Conditional Distribution 2

25

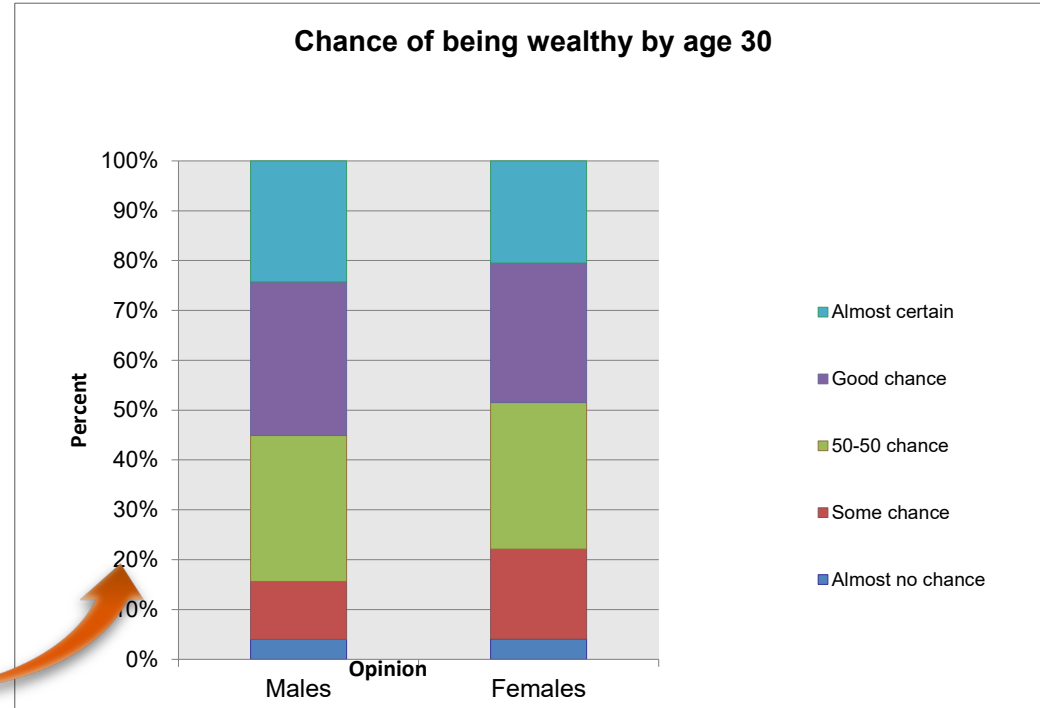
Young adults by gender and opinion of getting rich

	Female	Male	Total
Almost no chance	96	98	194
Some chance, but probably not	426	286	712
A 50–50 chance	696	720	1416
A good chance	663	758	1421
Almost certain	486	597	1083
Total	2367	2459	4826

Calculate the conditional distribution of opinion among males.

Examine the relationship between gender and opinion.

Response	Male	Female
Almost no chance	$98/2459 = 4.0\%$	$96/2367 = 4.1\%$
Some chance	$286/2459 = 11.6\%$	$426/2367 = 18.0\%$
A 50–50 chance	$720/2459 = 29.3\%$	$696/2367 = 29.4\%$
A good chance	$758/2459 = 30.8\%$	$663/2367 = 28.0\%$
Almost certain	$597/2459 = 24.3\%$	$486/2367 = 20.5\%$



Another example

26



	Sport Utility Vehicle (SUV)	Sports Car	Totals
male	21	39	60
female	135	45	180
Totals	156	84	240

ADD

ADD

Row Totals

Column Totals

Another example

27

Joint Frequency	Sport Utility Vehicle (SUV)	Sports Car	Totals	Marginal Frequency
	male	female		
male	21	39	60	Marginal Frequency
female	135	45	180	
Totals	156	84	240	

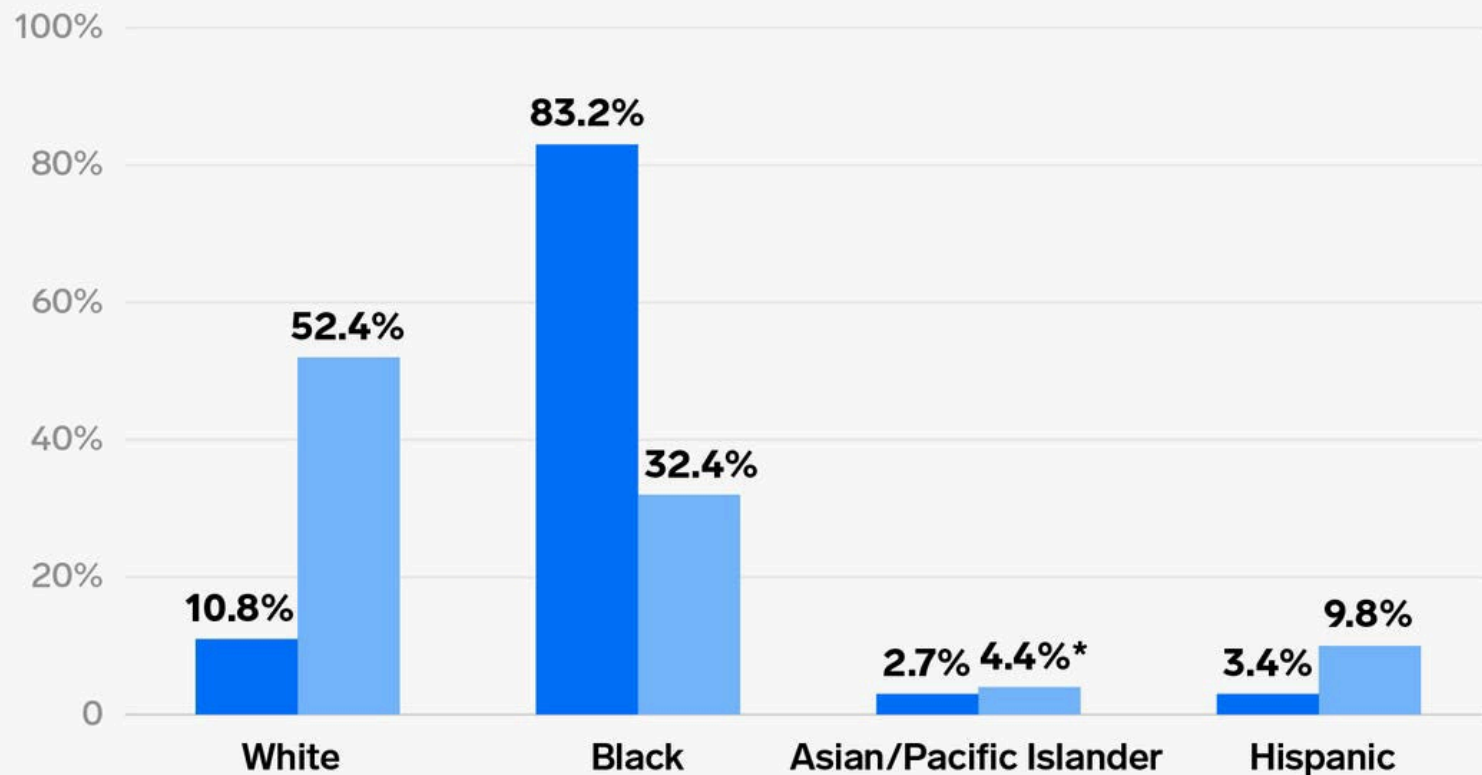
Another example

28

		Row Relative Frequencies - Divide male row by 60. Divide female row by 180. Divide Totals row by 240.	
Conditional relative frequencies		Sport Utility Vehicle (SUV)	Sports Car
male	$\frac{21}{60} = 0.35$	$\frac{39}{60} = 0.65$	$\frac{60}{60} = 1.00$
female	$\frac{135}{180} = 0.75$	$\frac{45}{180} = 0.25$	$\frac{180}{180} = 1.00$
Totals	$\frac{156}{240} = 0.65$	$\frac{84}{240} = 0.35$	$\frac{240}{240} = 1.00$

Georgia state COVID-19 hospitalizations, by race

■ % of COVID-19 hospitalizations in Georgia ■ % of population in Georgia



Simpson's Paradox 1

30

Consider the acceptance rates for the following groups of men and women who applied to a college.

Counts	Accepted	Not accepted	Total
Men	198	162	360
Women	88	112	200
Total	286	274	560

Percents	Accepted	Not accepted
Men	55%	45%
Women	44%	56%

A higher percentage of men were accepted.

Discussion: Is there evidence of discrimination?

Simpson's Paradox 2

31

Consider the acceptance rates when broken down by type of school.

BUSINESS SCHOOL

Counts	Accepted	Not accepted	Total
Men	18	102	120
Women	24	96	120
Total	42	198	240

Percents	Accepted	Not accepted
Men	15%	85%
Women	20%	80%

ART SCHOOL

Counts	Accepted	Not accepted	Total
Men	180	60	240
Women	64	16	80
Total	244	76	320

Percents	Accepted	Not accepted
Men	75%	25%
Women	80%	20%

Simpson's Paradox 3

32

An association or comparison that holds for several groups can reverse direction when the data are merged to a single group. This reversal is called **Simpson's paradox**.

Potential cause: When studying the relationship between two variables, there might exist a **lurking variable** that creates a reversal in the direction of the relationship when the lurking variable is ignored as opposed to the direction of the relationship when the lurking variable is considered.

The lurking variable creates subgroups, and failure to take these subgroups into consideration can lead to misleading conclusions regarding the association between the two variables.

Simpson's Paradox 4

33

- ✓ **Lurking variable:** Applications were split between the Business School (240) and the Art School (320).

Within each school a higher percentage of women were accepted than men.

This is an example of **Simpson's paradox**.

When the lurking variable (Type of School: Business or Art) is ignored, the data seem to suggest lower acceptance rate of women.

However, when the type of school is considered, the association is reversed.